

Machine Learning Notes

1. Introduction to Machine Learning

□ What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make decisions or predictions **without being explicitly programmed**.

Instead of writing fixed rules, we provide data, and the system learns from it.

□ Why Machine Learning?

Machine Learning is used because:

- Data is growing rapidly
 - Manual programming is not scalable
 - Complex patterns cannot be coded manually
 - Better accuracy improves over time
 - Automation of decision-making
-

□ AI vs ML

Artificial Intelligence (AI)	Machine Learning (ML)
Broad field of smart systems	Subset of AI
Focus on intelligence	Focus on learning from data
Can include rules + logic	Learns from datasets
Example: Robots, Chatbots	Example: Spam detection

□ Applications of ML

- Email spam detection
 - Recommendation systems (Netflix, YouTube)
 - Fraud detection
 - Healthcare diagnosis
 - Self-driving cars
 - Speech recognition
-

❑ Traditional Programming vs Machine Learning

Traditional Programming	Machine Learning
Input + Rules → Output	Input + Output → Model
Rules are fixed	Rules are learned
Hard to scale	Easily improves with data
Example: Calculator	Example: Face recognition

❑ Machine Learning Workflow

1. Data Collection
 2. Data Preprocessing (cleaning)
 3. Model Selection
 4. Training the model
 5. Testing the model
 6. Evaluation
 7. Deployment
-

❑ Types of Machine Learning

- Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
-

2. Types of Machine Learning

1. Supervised Learning

❑ Concept

In supervised learning, the model is trained using **labeled data**, meaning both input and correct output are given.

The model learns mapping:

❑ Input → Output

☐ **Key Concepts**

☐ **Labeled Data**

Data that contains both input features and correct answers.

☐ **Input Features**

Independent variables used for prediction.

☐ **Target Variable**

Output or result we want to predict.

☐ **Prediction**

Model output after training.

☐ **Training Process**

Model learns patterns from labeled dataset.

Types of Supervised Learning

1. Regression

- Predicts **continuous values**
 - Example:
 - House price prediction
 - Temperature prediction
-

2. Classification

- Predicts **categories or labels**
 - Example:
 - Spam or Not Spam
 - Disease detection (Yes/No)
-

2. Unsupervised Learning

□ **Concept**

In unsupervised learning, the model is trained using **unlabeled data** (no output given).

It finds hidden patterns in data.

□ **Key Concepts**

□ **Unlabeled Data**

Only input features are given, no correct output.

□ **Pattern Discovery**

Finding hidden structures in data.

□ **Similarity Detection**

Grouping similar data points.

Types of Unsupervised Learning

1. Clustering

- Grouping similar data
 - Example:
 - Customer segmentation
-

2. Dimensionality Reduction

- Reducing number of features
 - Keeps important information
 - Example:
 - PCA (Principal Component Analysis)
-

3. Association Learning

- Finds relationships between items

- Example:
 - “People who buy bread also buy butter”
-

3. Reinforcement Learning (Basic)

☐ **Concept**

An agent learns by interacting with an environment and receiving rewards or penalties.

☐ **Key Concepts**

☐ **Agent**

Learner or decision maker.

☐ **Environment**

The world in which agent operates.

☐ **Reward**

Feedback signal (positive or negative).

☐ **Actions**

Choices made by the agent.

☐ **Policy**

Strategy used by agent to choose actions.

3. Machine Learning Workflow

☐ **Complete Workflow Overview**

Machine Learning workflow is the step-by-step process used to build and deploy an ML model.

1. Problem Statement

- Define the objective clearly
 - Understand what needs to be predicted or solved
 - Example:
 - Predict house price
 - Classify emails as spam or not spam
-

2. Data Collection

- Gather relevant data from sources:
 - Databases
 - APIs
 - Files (CSV, Excel)
 - Sensors / Web scraping

☐ Data quality directly affects model performance.

3. Data Cleaning

- Remove incorrect or missing data
 - Fix inconsistencies
 - Handle noisy data
-

4. Data Preprocessing

- Convert raw data into usable format
 - Prepare data for ML algorithms
-

5. Feature Engineering

- Creating or modifying features to improve model performance
 - Examples:
 - Extracting year from date
 - Creating ratios or new variables
-

6. Model Selection

- Choose suitable algorithm:
 - Linear Regression
 - Decision Tree
 - KNN
 - SVM
-

7. Model Training

- Feed training data to model
 - Model learns patterns and relationships
-

8. Model Evaluation

- Check performance using metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-score
-

9. Hyperparameter Tuning

- Improve model performance by adjusting parameters
 - Techniques:
 - Grid Search
 - Random Search
-

10. Deployment

- Deploy model into real-world system:
 - Web apps
 - Mobile apps
 - APIs
-

11. Monitoring

- Continuously check model performance

- Detect:
 - Data drift
 - Performance drop
 - Retrain if needed
-

Workflow Summary

Problem → Data → Cleaning → Preprocessing → Features → Model → Training → Evaluation → Tuning → Deployment → Monitoring

4. Data Preprocessing

Data preprocessing is the process of cleaning and transforming raw data into a format suitable for machine learning.

1. Handling Missing Values

Missing data can reduce model accuracy.

Methods:

- Remove rows/columns
 - Fill with mean/median/mode
 - Use forward/backward fill
-

2. Removing Duplicate Data

- Eliminates repeated records
 - Prevents bias in training
-

3. Handling Outliers

Outliers are extreme values that affect model performance.

Methods:

- Remove outliers

- Cap values (clipping)
 - Use IQR method or Z-score
-

4. Encoding Categorical Data

Convert categorical data into numeric form.

☐ Label Encoding

- Assigns numbers to categories
 - Example:
 - Red = 0, Blue = 1, Green = 2
 - Best for ordinal data
-

☐ One Hot Encoding

- Creates binary columns
- Example:

Color	Red	Blue	Green
Red	1	0	0

- Best for nominal data
-

☐ 5. Feature Scaling

Used to bring all features to same scale.

☐ Standardization

- Mean = 0, Std = 1
- Formula:
 - $(x - \text{mean}) / \text{std}$

✓ Used when data has outliers or normal distribution

☐ **Normalization**

- Scales values between 0 and 1
- Formula:
 - $(x - \min) / (\max - \min)$

✓ Used for neural networks & distance-based models

6. Train-Test Split

- Split dataset into:
 - Training data (70–80%)
 - Testing data (20–30%)

Purpose:

- Train model on training data
- Evaluate on unseen data

5. Exploratory Data Analysis (EDA)

☐ **What is EDA?**

Exploratory Data Analysis (EDA) is the process of analyzing datasets to:

- Understand data structure
 - Find patterns
 - Detect errors, outliers, and relationships
 - Summarize important insights before modeling
-

1. Understanding Dataset

- Check shape (rows, columns)
- Identify data types (numeric, categorical)
- View sample records
- Understand features and target variable

☐ Helps in getting overall idea of dataset

2. Statistical Summary

Used to understand data distribution.

Key measures:

- Mean
- Median
- Mode
- Standard deviation
- Min & Max values

☐ Helps detect data imbalance and spread

3. Correlation Analysis

Measures relationship between variables.

- Range: -1 to +1
 - +1 → strong positive correlation
 - -1 → strong negative correlation
 - 0 → no relationship

☐ Helps in feature selection

4. Outlier Detection

Outliers are extreme values that differ from rest of data.

Methods:

- Box plot method (IQR)
- Z-score method
- Scatter plots

☐ Outliers can reduce model accuracy

5. Pattern Analysis

- Identifying trends in data

- Understanding relationships between variables
 - Finding hidden structures
-

Visualization Concepts (EDA Tools)

1. Histogram

- Shows distribution of numerical data
- Helps understand frequency

☐ Example: Age distribution

2. Bar Plot

- Used for categorical data
- Compares values between categories

☐ Example: Number of students per branch

3. Box Plot

- Shows:
 - Median
 - Quartiles
 - Outliers

☐ Best for detecting outliers

4. Scatter Plot

- Shows relationship between two variables
- Helps detect correlation

☐ Example: Height vs Weight

5. Heatmap

- Visual representation of correlation matrix
- Dark/light colors show strength of relationship

☐ Very useful for feature selection

☐ EDA Summary

- Understand data
 - Find patterns
 - Detect outliers
 - Use visualization tools
 - Prepare data for ML modeling
-

6. Feature Engineering

☐ What is Feature Engineering?

Feature Engineering is the process of creating, selecting, or transforming features to improve model performance.

☐ It is one of the most important steps in ML

1. Feature Selection

- Choosing important features
- Removing irrelevant or redundant features

Benefits:

- Improves accuracy
 - Reduces overfitting
 - Reduces computation cost
-

2. Feature Extraction

- Creating new features from existing data

- Reduces complexity

□ Example:

- Extracting year, month from date
-

3. Feature Transformation

- Changing data format or scale
 - Includes normalization, scaling, encoding
-

4. Polynomial Features

- Creates new features using combinations of existing features

Example:

- $x \rightarrow x^2, x^3, xy$

□ Helps in capturing non-linear relationships

5. Handling Imbalanced Data

When one class dominates others.

Techniques:

- Oversampling (SMOTE)
- Undersampling
- Class weighting

□ Important for classification problems

6. Dimensionality Reduction

- Reducing number of features while keeping important information

Methods:

- PCA (Principal Component Analysis)
- LDA (Linear Discriminant Analysis)

□ Benefits:

- Faster training
- Less overfitting
- Easier visualization

7. Regression Algorithms

□ **What is Regression?**

Regression is a supervised learning technique used to predict **continuous numerical values** like:

- House price
 - Salary
 - Temperature
-

1. Linear Regression

□ **Concept**

Linear Regression finds a **linear relationship** between input (X) and output (Y).

□ Equation:

$$Y = mx + c$$

□ **Regression Line**

- Line that represents predicted values
 - Shows relationship between X and Y
-

□ **Best Fit Line**

- Line with minimum error
 - Closest to all data points
 - Minimizes distance between actual and predicted values
-

☐ **Cost Function**

Measures error in prediction.

☐ Formula idea:

- Difference between actual and predicted values

✓ Goal: Minimize cost function

☐ **Gradient Descent**

Optimization algorithm used to reduce cost function.

Working:

- Starts with random values
 - Updates parameters step-by-step
 - Moves toward minimum error
-

Types of Linear Regression

1. Simple Linear Regression

- One input (X)
- One output (Y)

☐ Example:

Salary = Experience

2. Multiple Linear Regression

- Multiple inputs
- One output

☐ Example:

Salary = Experience + Age + Skills

☐ **Assumptions of Linear Regression**

1. Linearity

- Relationship between X and Y is linear

2. Independence

- Observations are independent

3. Homoscedasticity

- Constant variance of errors
-

☐ **Evaluation Metrics**

☐ **MAE (Mean Absolute Error)**

- Average absolute difference between actual and predicted values
-

☐ **MSE (Mean Squared Error)**

- Squares the errors
 - Penalizes large errors more
-

☐ **RMSE (Root Mean Squared Error)**

- Square root of MSE
 - Gives error in original units
-

☐ **R² Score**

- Measures accuracy of model
 - Range: 0 to 1
 - 1 → perfect model
 - 0 → poor model
-

2. Polynomial Regression

☐ Concept

Used when data is **non-linear**

- ☐ Extends linear regression using powers of X
-

☐ Non-linear Relationships

- Data not forming straight line
 - Curved patterns
-

☐ Polynomial Features

- Converts X into X^2 , X^3 , etc.

☐ Example:

$$Y = a + bX + cX^2$$

3. Decision Tree Regression

☐ Concept

Uses tree structure to make predictions.

☐ Tree Splitting

- Dataset is split into branches
 - Based on conditions
 - Continues until final prediction
-

☐ Entropy

- Measures uncertainty in data
- Higher entropy = more disorder

☐ **Gini Index**

- Measures impurity of dataset
- Lower value = better split

☐ Used for selecting best split

4. Random Forest Regression

☐ **Concept**

- Ensemble method using multiple decision trees
 - Final output = average of all trees
-

☐ **Ensemble Learning**

Combining multiple models to improve accuracy.

☐ **Multiple Decision Trees**

- Each tree trained on random data
 - Reduces overfitting
-

☐ **Bagging (Bootstrap Aggregation)**

- Random sampling with replacement
 - Builds multiple models
 - Combines results
-

5. Support Vector Regression (SVR)

☐ **Concept**

SVR tries to fit the **best line within a margin of tolerance**

☐ **Hyperplane**

- Decision boundary used to predict values
-

☐ **Margin**

- Allowed error range around prediction line
-

☐ **Kernel Functions**

Used to handle non-linear data by transforming it into higher dimensions.

Common kernels:

- Linear Kernel
- Polynomial Kernel
- RBF (Radial Basis Function)

8. Classification Algorithms

☐ **What is Classification?**

Classification is a supervised learning technique used to predict **categories or labels**.

☐ **Output is discrete:**

- Yes / No
 - Spam / Not Spam
 - Disease / No Disease
-

1. Logistic Regression

☐ **Concept**

Logistic Regression is used for **binary classification problems**.

It predicts probability using a sigmoid function.

☐ Sigmoid Function

Converts any value into range **0 to 1**.

☐ Formula:

- Output = probability

✓ If value $> 0.5 \rightarrow$ Class 1

✓ If value $< 0.5 \rightarrow$ Class 0

☐ Binary Classification

- Two classes only
 - Example:
 - Pass / Fail
 - Spam / Not Spam
-

☐ Probability Prediction

- Model outputs probability instead of direct class
 - Helps in decision-making
-

2. K-Nearest Neighbors (KNN)

☐ Concept

KNN classifies data based on **nearest neighbors**.

☐ Distance Metrics

☐ Euclidean Distance

Most commonly used distance formula:

- ☐ Measures straight-line distance between points

☐ **Choosing K Value**

- K = number of neighbors
- Small K → noisy results
- Large K → smoother results

✓ Best K is chosen by testing accuracy

3. Support Vector Machine (SVM)

☐ **Concept**

SVM finds the **best separating boundary (hyperplane)** between classes.

☐ **Hyperplane**

- Decision boundary that separates classes
 - Example: line in 2D, plane in 3D
-

☐ **Support Vectors**

- Important data points closest to hyperplane
 - They define margin
-

☐ **Kernel Trick**

Used to handle non-linear data by transforming it into higher dimensions.

☐ **Types:**

- Linear Kernel
 - Polynomial Kernel
 - RBF Kernel
-

4. Naive Bayes

☐ Concept

Naive Bayes is a probabilistic classifier based on **Bayes Theorem**.

☐ Bayes Theorem

Calculates probability based on prior knowledge and evidence.

☐ Conditional Probability

Probability of event A given B.

☐ Independence Assumption

Assumes all features are independent (simplified assumption).

✓ Makes computation fast and efficient

5. Decision Tree Classifier

☐ Concept

Tree-based model used for classification using rules.

☐ Entropy

- Measures impurity or disorder in dataset
-

☐ Information Gain

- Reduction in entropy after split
- Used to choose best feature

☐ Gini Impurity

- Measures how often incorrect classification happens
 - Lower value = better split
-

6. Random Forest Classifier

☐ Concept

Random Forest is an **ensemble method** using multiple decision trees.

☐ Multiple Trees

- Each tree gives prediction
 - Final output is combined result
-

☐ Voting Mechanism

- Majority voting used for classification
 - Most voted class is final output
-

☐ Ensemble Learning

Combining multiple models to improve accuracy and reduce overfitting.

☐ Ensemble Learning Techniques

☐ Bagging (Bootstrap Aggregation)

- Multiple models trained on random samples
- Reduces variance
- Example: Random Forest

☐ **Boosting**

- Models trained sequentially
 - Each model improves previous errors
-

☐ **AdaBoost**

- Focuses on misclassified data
 - Increases weight of difficult samples
-

☐ **Gradient Boosting**

- Optimizes errors using gradient descent
 - Builds strong model step-by-step
-

☐ **XGBoost (Basic)**

- Advanced boosting technique
- Fast and highly accurate
- Used in competitions (Kaggle)

9. Clustering Algorithms

☐ **What is Clustering?**

Clustering is an **unsupervised learning technique** used to group similar data points together without labels.

- ☐ Goal: Find hidden patterns in data
-

1. K-Means Clustering

☐ **Concept**

K-Means divides data into **K clusters** based on similarity.

☐ Centroids

- Center point of each cluster
 - Updated iteratively until optimal position is found
-

☐ Distance Calculation

- Measures similarity between points and centroids
 - Common method: Euclidean distance
-

☐ Elbow Method

- Used to find optimal number of clusters (K)
 - Plot WCSS vs K
 - “Elbow point” = best K value
-

☐ Inertia

- Sum of squared distances between points and centroid
 - Lower inertia → better clustering
-

2. Hierarchical Clustering

☐ Concept

Builds clusters in a **tree-like structure**

☐ Dendrogram

- Tree diagram showing cluster formation
 - Helps decide number of clusters
-

☐ **Agglomerative Clustering**

- Bottom-up approach
 - Starts with individual points
 - Merges closest clusters step-by-step
-

3. DBSCAN (Density-Based Clustering)

☐ **Concept**

Groups data based on **density of points**

☐ **Core Points**

- Points with many neighbors
 - Form main cluster centers
-

☐ **Noise Points**

- Outliers
 - Do not belong to any cluster
-

☐ **Key Idea**

- Dense regions → clusters
 - Sparse regions → noise
-

☐ **Clustering Summary**

☐ **K-Means:**

- Centroid-based
- Uses Elbow method
- Minimizes inertia

☐ **Hierarchical:**

- Tree structure
- Dendrogram
- Agglomerative approach

☐ **DBSCAN:**

- Density-based
 - Handles outliers well
 - Finds irregular clusters
-

10. Dimensionality Reduction

☐ **What is Dimensionality Reduction?**

Process of reducing number of features while preserving important information.

☐ **Helps in:**

- Faster computation
 - Less overfitting
 - Better visualization
-

☐ **Principal Component Analysis (PCA)**

☐ **Concept**

PCA transforms data into a **new set of variables (principal components)**.

☐ **Variance**

- Measures spread of data
 - PCA keeps components with highest variance
-

☐ **Principal Components**

- New features created from original features
- Ranked by importance

☐ **Feature Reduction**

- Reduces number of features
 - Keeps maximum information with fewer dimensions
-

☐ **PCA Working Idea (Simple)**

1. Standardize data
2. Find covariance matrix
3. Compute eigenvalues & eigenvectors
4. Select top components
5. Transform data

11. Association Rule Learning

☐ **What is Association Rule Learning?**

Association Rule Learning is an **unsupervised learning technique** used to find relationships between items in large datasets.

- ☐ Goal: Discover “IF–THEN” patterns
-

☐ **Market Basket Analysis**

Used in retail to find products frequently bought together.

- ☐ Example:

- Bread → Butter
- Milk → Bread

Used by:

- Amazon
 - Walmart
 - Online shopping platforms
-

☐ **Apriori Algorithm**

☐ **Concept**

Apriori algorithm finds frequent itemsets and generates association rules.

☐ Works step-by-step:

1. Find frequent items
 2. Generate combinations
 3. Build rules
-

☐ **Key Metrics**

1. Support

Measures how frequently an item appears in dataset.

☐ Formula idea:

- $\text{Support} = (\text{Transactions containing item}) / (\text{Total transactions})$

✓ High support = commonly purchased item

2. Confidence

Measures how often rule is correct.

☐ Example:

IF Bread $\rightarrow$ Butter

✓ Confidence = probability that butter is bought when bread is bought

3. Lift

Measures strength of association between items.

☐ Interpretation:

- $\text{Lift} > 1 \rightarrow$ strong relationship

- Lift = 1 → independent
 - Lift < 1 → negative relationship
-

☐ Quick Summary

Metric	Meaning
Support	Frequency
Confidence	Rule accuracy
Lift	Strength of relationship

12. Model Evaluation

☐ What is Model Evaluation?

Model evaluation checks how well a machine learning model performs on unseen data.

☐ Regression Metrics

☐ MAE (Mean Absolute Error)

- Average of absolute errors
 - Simple and interpretable
-

☐ MSE (Mean Squared Error)

- Squares errors
 - Penalizes large mistakes
-

☐ RMSE (Root Mean Squared Error)

- Square root of MSE
 - Gives error in original units
-

☐ R² Score

- Measures how well model explains variance
- Range: 0 to 1

✓ 1 = perfect model

☐ **Classification Metrics**

☐ **Accuracy**

- Overall correctness of model
-

☐ **Precision**

- How many predicted positives are actually correct
-

☐ **Recall**

- How many actual positives are correctly predicted
-

☐ **F1 Score**

- Balance between Precision and Recall

☐ Useful when dataset is imbalanced

☐ **Confusion Matrix**

Shows prediction results:

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

☐ **Advanced Evaluation**

☐ **ROC Curve**

- Graph between True Positive Rate vs False Positive Rate
 - Shows classification performance
-

☐ **AUC Score**

- Area under ROC curve
 - Higher AUC = better model
-

☐ **Cross Validation**

Technique to evaluate model multiple times using different data splits.

☐ **K-Fold Cross Validation**

- Data is split into K parts
 - Model trained K times
 - Each fold used as test once
-

☐ **Bias vs Variance**

☐ **Bias**

- Error due to overly simple model
 - Leads to underfitting
-

☐ **Variance**

- Error due to complex model
- Leads to overfitting

☐ **Tradeoff**

- Balance needed between bias and variance for best model
-

Overfitting vs Underfitting

☐ **Overfitting**

- Model learns training data too well
 - Poor performance on test data
 - High variance
-

☐ **Underfitting**

- Model is too simple
 - Poor performance on both training and test data
 - High bias
-

13. Hyperparameter Tuning

☐ **What is Hyperparameter Tuning?**

Hyperparameter tuning is the process of selecting the **best parameters for a machine learning model** to improve performance.

- ☐ These parameters are not learned from data; they are set manually.
-

1. Grid Search CV

☐ **Concept**

Grid Search tries **all possible combinations** of hyperparameters.

☐ **Working**

- Define parameter grid
 - Try every combination
 - Select best performing model
-

☐ **Cross Validation (CV)**

- Model is evaluated multiple times on different data splits
 - Gives reliable performance estimate
-

☐ **Advantage**

- ✓ Finds best combination

☐ **Disadvantage**

- ☐ Very slow for large datasets
-

2. Random Search CV

☐ **Concept**

Random Search selects **random combinations** of hyperparameters.

☐ **Working**

- Randomly picks parameter values
 - Evaluates model
 - Repeats many times
-

☐ **Advantage**

- ✓ Faster than Grid Search
 - ✓ Works well for large search space
-

☐ **Disadvantage**

- ☐ May miss best combination
-

3. Cross Validation (CV)

☐ **Concept**

Cross Validation is used to evaluate model performance more reliably.

☐ **Why CV is used?**

- Avoids overfitting on single train-test split
 - Gives stable accuracy
-

☐ **Common Type:**

- K-Fold Cross Validation

- ☐ Data is split into K parts and model is trained K times.
-

☐ **Quick Revision**

Method	Key Idea
Grid Search	Try all combinations
Random Search	Try random combinations
Cross Validation	Evaluate model on multiple splits

14. Regularization

☐ **What is Regularization?**

Regularization is a technique used to **prevent overfitting** by adding penalty to model complexity.

☐ **Overfitting Prevention**

Overfitting happens when:

- Model learns training data too well
- Performs poorly on test data

✓ Regularization helps simplify model

1. L1 Regularization (Lasso)

☐ **Concept**

L1 adds **absolute value penalty** to coefficients.

☐ **Effect**

- Some weights become **zero**
 - Performs **feature selection**
-

☐ **Formula Idea:**

- $\text{Cost} + \lambda (\text{sum of absolute weights})$
-

☐ **Advantages**

- ✓ Feature selection
 - ✓ Simple model
-

☐ **Disadvantages**

- ☐ Can remove important features sometimes
-

2. L2 Regularization (Ridge)

☐ Concept

L2 adds **squared penalty** to coefficients.

☐ Effect

- Reduces weight values
 - Does not make weights zero
-

☐ Formula Idea:

- $\text{Cost} + \lambda (\text{sum of squared weights})$
-

☐ Advantages

- ✓ Prevents overfitting
 - ✓ Stable model
-

☐ Disadvantages

- ☐ No feature selection
-

3. ElasticNet

☐ Concept

ElasticNet is a **combination of L1 + L2 regularization**.

☐ Working

- Uses both Lasso + Ridge penalties
- Balances feature selection and stability

☐ Advantages

- ✓ Best of both L1 and L2
- ✓ Works well with correlated features

15. Time Series Basics

☐ What is Time Series?

Time series data is data collected over **time in sequential order**.

☐ Example:

- Stock prices ☐
 - Weather data ☐
 - Sales data ☐
-

1. Trend

☐ Concept

Trend shows the **long-term movement** in data.

☐ Types:

- Upward trend ☐
- Downward trend ☐
- No trend

- ✓ Example: Increasing sales over years
-

2. Seasonality

☐ Concept

Seasonality refers to **repeating patterns at fixed intervals**.

☐ Example:

- Ice cream sales increase in summer

- Sales spike during festivals

✓ Happens due to seasons, holidays, cycles

3. Forecasting Basics

☐ Concept

Forecasting means **predicting future values using past data.**

☐ Steps:

1. Analyze historical data
 2. Identify trend & seasonality
 3. Build model
 4. Predict future values
-

4. Moving Average

☐ Concept

Moving average smooths data by taking **average of past values.**

☐ Purpose:

- Remove noise
 - Identify trend clearly
-

☐ Types:

- Simple Moving Average (SMA)
 - Weighted Moving Average (WMA)
-

☐ Quick Revision

□ **Time Series:**

- Data in time order
- Used for forecasting

□ **Components:**

- Trend → long-term direction
 - Seasonality → repeating pattern
 - Forecasting → future prediction
 - Moving average → smoothing
-

16. Recommendation Systems

□ **What is a Recommendation System?**

A system that suggests relevant items to users based on preferences or behavior.

□ **Example:**

- Netflix □ suggestions
 - Amazon □ product recommendations
 - YouTube □ videos
-

1. Content-Based Filtering

□ **Concept**

Recommends items similar to what the user liked before.

□ **Working:**

- Analyze item features
 - Match similar items
-

□ **Example:**

If you like action movies → recommend other action movies

☐ **Advantage:**

- ✓ Personalized recommendations

☐ **Disadvantage:**

- ☐ Limited variety (no new interests)
-

2. Collaborative Filtering

☐ **Concept**

Recommends items based on **other users' behavior**.

☐ **Working:**

- Find similar users
 - Recommend what similar users liked
-

☐ **Types:**

- User-based filtering
 - Item-based filtering
-

☐ **Example:**

If many users who liked A also liked B → recommend B

☐ **Advantage:**

- ✓ Discovers new interests

☐ **Disadvantage:**

- ☐ Cold start problem (new users/items)
-

3. Hybrid Recommendation System

- ☐ **Concept**

Combines **content-based** + **collaborative filtering**.

- ☐ **Working:**

- Uses both user preferences and behavior patterns
-

- ☐ **Advantage:**

- ✓ More accurate recommendations
 - ✓ Solves limitations of both methods
-

- ☐ **Example:**

Netflix + Amazon combined recommendation logic